

ML-Based Music Genre Classification

Machine Learning in Practice Project

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1 Introduction

Both machine learning and music analysis frequently deal with the classification of music into various genres. People can typically distinguish between different genres of music with little effort, but teaching a machine to do the same is more challenging due to the complexity and abundance of features in audio signals. In this project, we developed machine learning models that can identify a song's genre based on two different input types: spectrogram images, which display the frequency changes over time, and numerical audio features extracted from music clips.

Our project's primary objective was to establish a comprehensive procedure that covers feature extraction, data preparation, model training, and performance assessment using accepted metrics. We used a kind of neural network known as a Multi-layer Perceptron (MLP) to test and compare a number of traditional machine learning models, including Support Vector Machines (SVM), Random Forests, and K-Nearest Neighbors (KNN). To observe how each model functions with various input types, we applied these models to the spectrogram images as well as the audio feature data (in table form).

The datasets utilized in this research originate from the widely recognized GTZAN dataset, made publicly accessible through Kaggle. The collection includes the following files and directories:

- `features_3_sec.csv`: a dataset comprising feature vectors extracted from three-second audio segments.
- `features_30_sec.csv`: resembling the prior dataset but derived from more extensive, thirty-second audio segments.
- `images_original`: a directory containing spectrogram images created from the audio clips; these images encapsulate spectral properties of entire thirty-second tracks.
- `genres_original`: a directory containing the original .wav audio files, organized into genre-specific subfolders. These raw audio files provide the foundational basis for both feature extraction and spectrogram generation, ensuring traceability and reproducibility of derived data representations.

Important preprocessing procedures like normalizing the data, lowering its dimensionality, and turning genre labels into numbers were all part of our project. Following that, we trained various models on each dataset and assessed their performance using metrics like F1-score, ROC curves, and Precision-Recall curves. These metrics enabled us to assess each machine learning model's performance across a range of data types.

This report explains not just the technical steps we took, but also our analysis of the results and what they mean. We discuss the reasons behind each choice we made and what we learned from the models during the process. To make things clearer and highlight the most important points, we've also included graphs and visuals throughout the report.

2 Dataset

The datasets utilized in this study are derived from the GTZAN genre collection, which is one of the most well-known and commonly used datasets for music genre classification. Each audio track in this dataset has a duration of 30 seconds and is categorized into one of ten predefined genres: blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, and rock. The full dataset is composed of 1000 tracks, with 100 tracks for each genre.

2.1 Engineered Audio Features

The datasets, namely `features_3_sec.csv` and `features_30_sec.csv`, represent numerical features extracted through the analysis of audio files in .wav format, stored within the `genres_original/` directory. These features were generated using the LibROSA library alongside comparable digital signal processing methods, providing a systematic framework for characterization. Each data entry corresponds to a segment of music, meticulously annotated with its associated genre label.

The `features_30_sec.csv` dataset comprises a single data row for each song, encapsulating a 30-second musical excerpt per track. In contrast, the `features_3_sec.csv` dataset divides each track into ten discrete 3-second segments, resulting in a dataset with ten times more rows to facilitate finer-grain analysis.

The extracted features are designed to encapsulate both perceptual and statistical attributes of the audio signals. Such features are particularly suited for integration into machine learning pipelines, as they serve as quantitative representations of the audio's underlying properties. A detailed outline of the principal feature categories is provided below:

- **spectral features:** they encompass crucial information that pertains to the frequency constituents of sound, which are instrumental in characterizing elements such as timbre, brightness, and texture.
 - *Spectral centroid:* this feature denotes the "center of mass" of the spectrum. Elevated values are indicative of more vivid or sharper sounds, commonly associated with genres like metal or pop, whereas diminished values are prevalent in more mellow genres such as classical music.
 - *Spectral rolloff:* this parameter defines the frequency beneath which a specified percentage (typically 85%) of the total spectral energy is situated. It assists in understanding how rapidly the energy in the signal diminishes, proving useful in differentiating percussive elements (characterized by a sharp cutoff) from harmonic content.
 - *Spectral bandwidth:* This feature measures the distribution of the spectrum around its centroid. A broader bandwidth signifies a more chaotic or intricate timbre.
 - *Spectral contrast:* this quantifies the discrepancy in energy between peaks and troughs within the spectrum, aiding in discerning the dynamic range within the frequency domain.
- **temporal features:** they elucidate the time-domain behavior of an audio signal, which is essential for comprehending aspects of rhythm and transient activity.
 - *Zero crossing rate:* this evaluates how frequently the signal changes sign or crosses the zero amplitude axis. It is usually elevated in sonorous or percussive sounds such as drums or rap music but lower in smoother signals, such as those produced by classical string instruments.
 - *Tempo:* this refers to the estimated beats per minute (BPM), which derives from the rhythmic patterns of the audio. Genres characterized by brisk tempos include dance music styles such as disco or EDM, while slower tempos may be characteristic of ballads or classical music.
- **Mel-Frequency Cepstral Coefficients (MFCCs):** these coefficients provide a model of the short-term power spectra of sounds on a nonlinear Mel scale, approximating human auditory perception. MFCCs are predominantly employed in both speech and music recognition. The lower coefficients capture overarching spectral shapes, whereas higher coefficients focus on subtle details, including texture and articulation. MFCCs are particularly adept at encoding timbre, aiding in the differentiation of instruments or production styles, for example, comparing acoustic to electric sounds or smooth versus distorted textures.
- **Chroma features (Pitch Class Profiles):** they quantify the intensity of each pitch class (C, C#, D, ..., B) over time, independent of octave variations. Such features facilitate the detection of harmonic components and enable genre differentiation based on chord progressions and scale usage, such as distinguishing jazz from metal.

- **Root Mean Square Energy (RMSE):** this measures the loudness or energy present within an audio signal. Elevated RMSE values may suggest intense or dense musical sections (e.g., rock choruses), whereas low values could indicate softer or ambient musical textures.
- **chromagram and fonal centroid features:** in more comprehensive datasets, additional features may include:
 - *Tonnetz*: it describes tonal relationships such as harmony and key, derived from harmonic content.
 - *Chroma CQT* or *Chroma CENS*: alternative chroma descriptors derived from various Fourier transforms.

The datasets are well-structured and free of null values, with all features being complete and numeric. This makes them readily usable for machine learning tasks without any need for imputation or correction. The class distribution is perfectly balanced, as each of the ten genres is equally represented. There are 100 samples per genre in the 30-second dataset and about 1000 per genre in the 3-second dataset. This balance eliminates the need for resampling techniques to address class imbalance. Additionally, it's worth noting that many features, particularly the MFCCs and chroma-related attributes, often exhibit high correlation with each other. This interdependence is a key factor in why dimensionality reduction techniques, such as Principal Component Analysis (PCA), are commonly used before classification. They enhance model efficiency and help reduce the risk of overfitting. Lastly, genre labels are encoded in the label column, mapping each instance to one of the predefined categories: blues, classical, country, disco, hip hop, jazz, metal, pop, reggae, or rock.

2.2 Spectrogram Images

The `images_original` folder comprises 1.000 PNG files, each depicting the spectrogram of a complete 30-second audio segment. These images are systematically arranged within subfolders, each named according to the corresponding genre labels. Each spectrogram illustrates the temporal frequency spectrum, generated via Short-Time Fourier Transform (STFT), where lighter areas signify greater energy or intensity at particular frequencies and times.

3 Method

The methodological framework for this project is structured to keep a strong emphasis on rigor and reproducibility. The selection of techniques for preprocessing, feature transformation, and classification was based on established best practices in machine learning and carefully tailored to suit the specific attributes of each dataset.

3.1 Preprocessing for tabular data

The initial step in processing the two CSV datasets, `features_3_sec.csv` and `features_30_sec.csv`, involved cleaning the data by removing columns that did not contribute meaningful discriminative information. Specifically, meta-data fields such as `'filename'` and `'length'`, as well as columns ending with `'_var'` representing variance values of audio descriptors, were excluded. While variance features can sometimes provide useful insights, they often introduce noise and multicollinearity. Furthermore, preliminary analyses indicated that their inclusion did not enhance classification performance.

Subsequently, the genre labels were transformed into numerical format using the `'LabelEncoder'` from `scikit-learn`. This process replaced string-based labels (e.g., rock, jazz) with integers ranging from 0 to 9. Such label encoding was essential to meet the requirements of machine learning models within `scikit-learn`, which operate on numerical target variables.

To facilitate optimal performance of distance-based classification algorithms such as Support Vector Machines (SVM) and k-Nearest Neighbors (KNN), all features were standardized using `'StandardScaler'`. This procedure ensured that each feature had a mean of zero and a standard deviation of one. Standardization was particularly critical due to the disparate scales of features, such as tempo and spectral bandwidth; unscaled features may otherwise disproportionately influence these models.

To address the issue of high dimensionality present in the feature sets, Principal Component Analysis (PCA) was employed for dimensionality reduction. In addition to mitigating computational complexity, PCA served to eliminate redundancy and noise in the feature space, thereby potentially enhancing model generalization. For both the 3-second and 30-second datasets, the first 29 principal components were retained, capturing the majority of the data's total variance.

After completing the preprocessing steps, each dataset was divided into training and testing subsets using a stratified split with an 80/20 ratio. Stratification ensured that the distribution of genre classes remained consistent across both subsets, thus preserving representativeness in the training and testing phases.

3.2 Preprocessing for image data

The spectrogram images stored in the `images_original` folder were each categorized into one of ten music genres and organized into corresponding subfolders. To prepare these images for analysis, we first converted them to grayscale. This reduced the data's dimensionality by removing the color channels while retaining critical frequency-time information, as the key spectral features remained intact in grayscale.

Next, each image was resized to a fixed dimension of 64×64 pixels, ensuring a consistent format necessary for batch processing and compatibility with traditional machine learning models. The resized images were then flattened into 4096-dimensional vectors by converting the 2D pixel grid into a single array. While this flattening step eliminated spatial relationships—features—it enabled us to work with standard classifiers such as Support Vector Machines (SVM) and Multi-Layer Perceptrons (MLP).

Following vectorization, the dataset underwent standardization with `StandardScaler` to normalize pixel intensity values, ensuring a mean of zero and a unit variance across all features.

Finally, the dataset was split into training and test subsets using an 80/20 stratified approach. This method preserved the balance of genres within both sets, ensuring reliable representation during model training and evaluation.

3.3 Model selection and hyperparameter tuning

For each tabular dataset, we trained and assessed the performance of four distinct classifiers: Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), and Multi-layer Perceptron (MLPClassifier). Each model was selected for its unique approach to learning and its potential effectiveness across different data domains. For the image dataset, instead, only the SVM and the MLPClassifier approach were used.

- **SVM**, known for its robust margin-based classification, performs particularly well in high-dimensional spaces. We utilized both radial basis function (RBF) and linear kernels, conducting a grid search to optimize C (regularization) and γ parameters.

- **Random Forest**, an ensemble learning method that leverages decision trees, excels at capturing non-linear patterns in structured data. For this classifier, we fine-tuned key parameters such as the number of estimators and the maximum tree depth.
- **KNN**, a non-parametric, instance-based algorithm, relies heavily on the selection of the distance metric and the number of neighbors (k). We experimented with multiple k values, including 3, 5, and 7.
- **MLPClassifier**, a type of feedforward neural network with at least one hidden layer, was deployed as a black-box model with minimal configuration. We opted for a single hidden layer containing 100 neurons, using default hyperparameters to align with the project goal of evaluating it without extensive tuning.

Hyperparameter optimization for all models, except the MLPClassifier, was carried out using GridSearchCV with 3-fold cross-validation. This systematic approach helped identify parameter configurations that balanced generalization performance while mitigating overfitting risks.

3.4 Evaluation strategy

To assess the performance of the model, we utilized a comprehensive array of evaluation metrics. The primary metric used was the weighted F1-score, which adjusts for class imbalance by calculating the F1-score for each class individually and averaging them based on the class support. This approach is particularly well-suited for multi-class classification tasks, where the balance between precision and recall can differ across labels.

Beyond F1-scores, ROC curves were plotted for each class using a one-versus-rest strategy, accompanied by the calculation of the Area Under the Curve (AUC). These curves highlight the trade-off between true positive and false positive rates, while the AUC provides a summarized measure of the classifier's confidence and ranking capabilities.

Additionally, precision-recall curves were generated to evaluate the model's ability to sustain high precision across varying levels of recall. This metric is especially crucial in fields where minimizing false positives holds significant importance.

Lastly, confusion matrices were created for each model to visualize patterns in misclassification across different categories. These matrices offered valuable insights into common errors, such as frequent misidentification of closely related genres like jazz and blues, and helped pinpoint areas in need of improvement within each model.

4 Experiments and results

This section provides a detailed account of the experimental procedures and results derived from applying machine learning classifiers to three distinct datasets: two engineered audio feature datasets—one based on 3-second segments and the other on 30-second segments—and a dataset consisting of spectrogram images. The analysis adhered to the requisite experimental framework, involving model training, hyperparameter optimization (with the exception of MLPClassifier), evaluation of performance metrics, and a comparative assessment of the outcomes.

For the engineered feature datasets, classification was performed using Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), and Multi-Layer Perceptron (MLPClassifier). In contrast, only SVM and MLP were utilized for the spectrogram images. The evaluation of all models relied on key metrics such as F1-score, Receiver Operating Characteristic (ROC), precision-recall curves, and their associated Area Under the Curve (AUC) values. Hyperparameter tuning was conducted through grid search for all classifiers except MLPClassifier, which was employed in its default configuration as a baseline model.

The 3-second feature dataset comprised short, segmented excerpts from the original audio tracks, resulting in a large training set. However, this approach introduced increased variance and a loss of contextual information, as individual fragments often failed to encapsulate the entire genre properties. Among classifiers, SVM produced the highest weighted F1-score of approximately 0.80. Random Forest followed with a score of 0.78, while MLPClassifier performed comparably with a score of 0.76 despite the absence of parameter tuning. KNN achieved a lower F1-score of around 0.75 and faced challenges in distinguishing genres with overlapping timbres or rhythmic characteristics. These findings were corroborated by the ROC and precision-recall curves. SVM demonstrated the highest AUC values, indicating strong inter-class separability, whereas KNN exhibited pronounced overlaps and weaker discrimination capabilities.

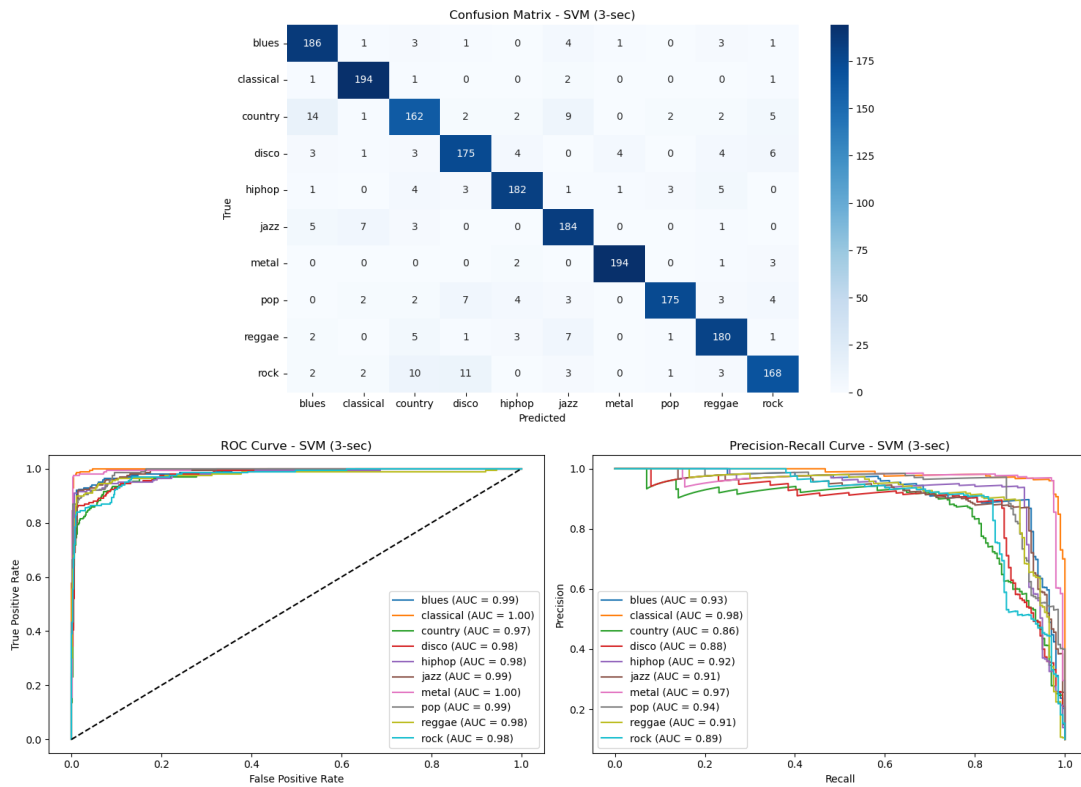


Figure 1: SVM evaluation on the 3-second feature dataset. Confusion matrix, ROC, and precision-recall curves show strong genre separation, with some overlap in similar styles.

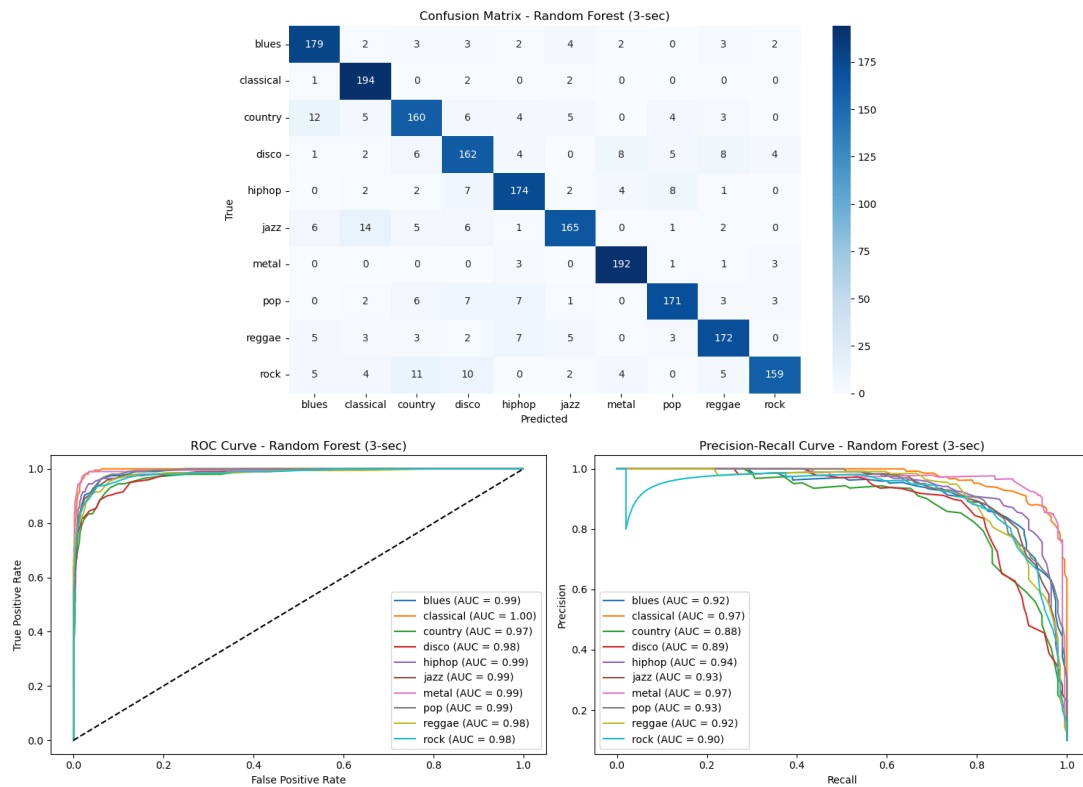


Figure 2: Random Forest results on the 3-second features. Consistent classification and high AUC across most genres, with minor confusion in mid-energy classes.

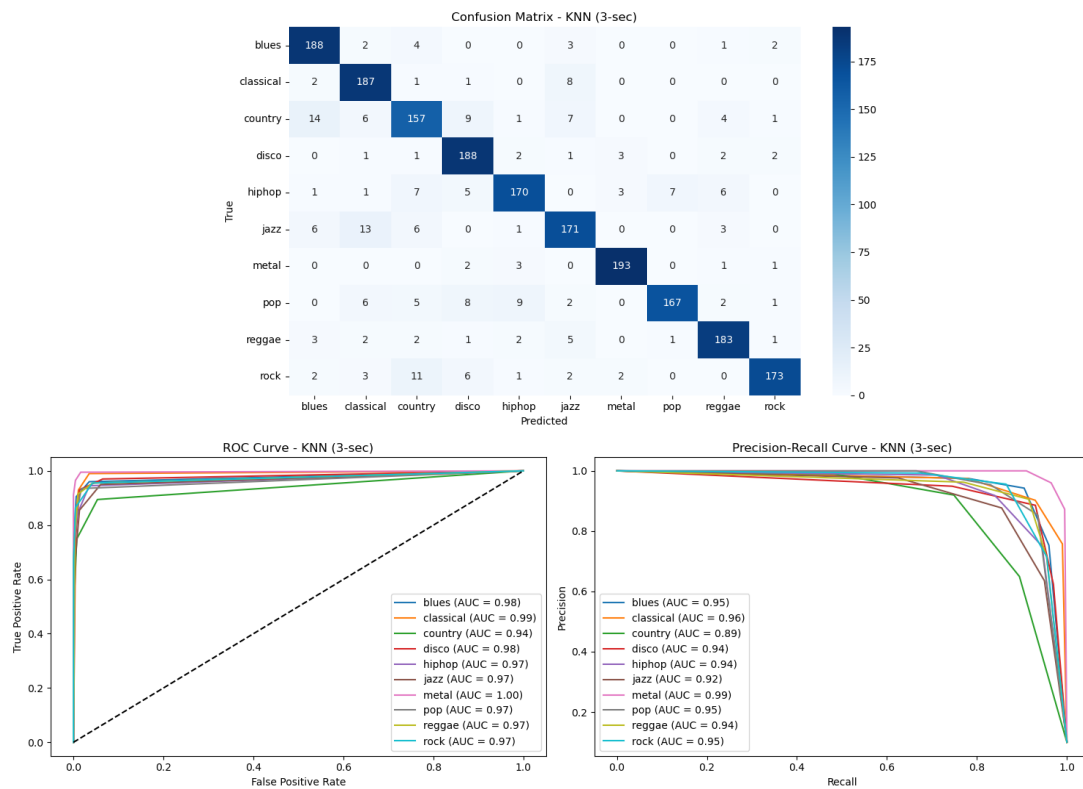


Figure 3: KNN performance on the 3-second dataset. Lower accuracy and more class overlap.

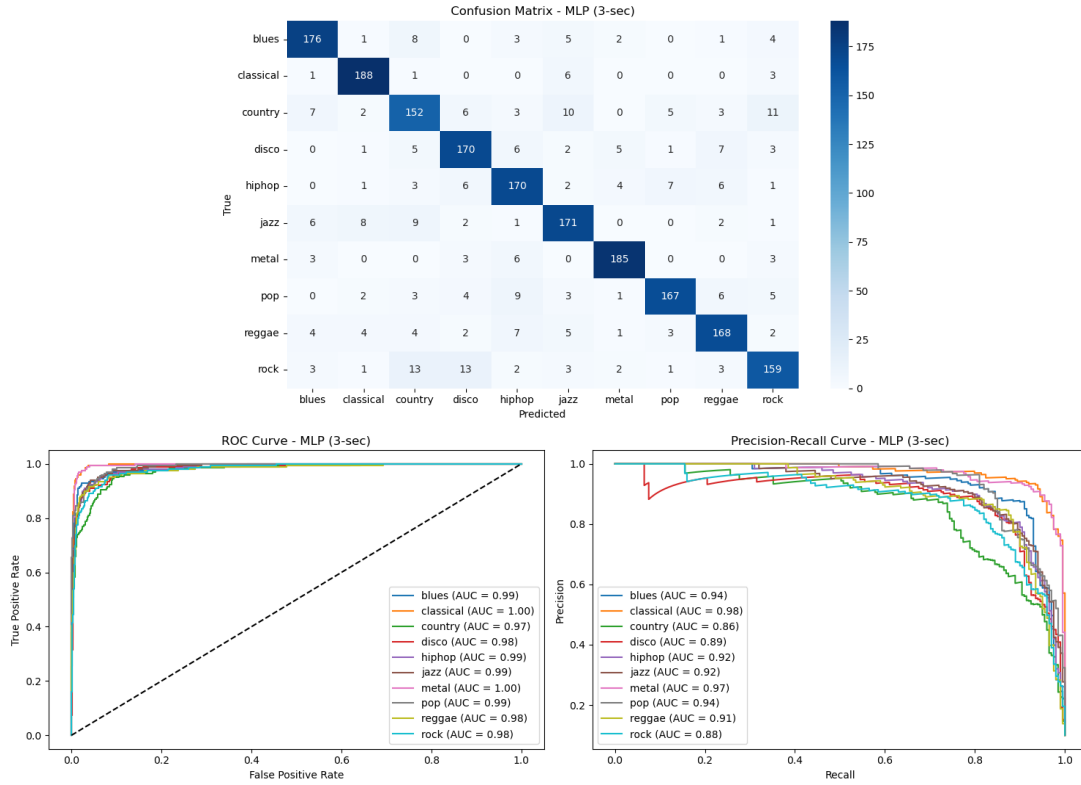


Figure 4: MLPClassifier on 3-second features. Competitive results with solid recall despite default parameters.

In the case of the 30-second feature dataset, longer track durations provided richer temporal and harmonic information, thereby enhancing classifier performance across all models. SVM again emerged as the top performer with an F1-score of 0.84, followed closely by MLPClassifier at 0.82 and Random Forest at 0.81. While KNN lagged behind with an F1-score of 0.78, it showed marked improvement compared to its performance on the 3-second dataset. The inclusion of longer audio segments captured more complete musical structures, leading to reduced ambiguity and improved classifier reliability. In line with these observations, the evaluation curves demonstrated better class separability and higher AUC values than those achieved with the 3-second dataset. SVM consistently outperformed other models in AUC scores across nearly all genres, with only minor reductions in performance for stylistically similar genres such as jazz and blues.

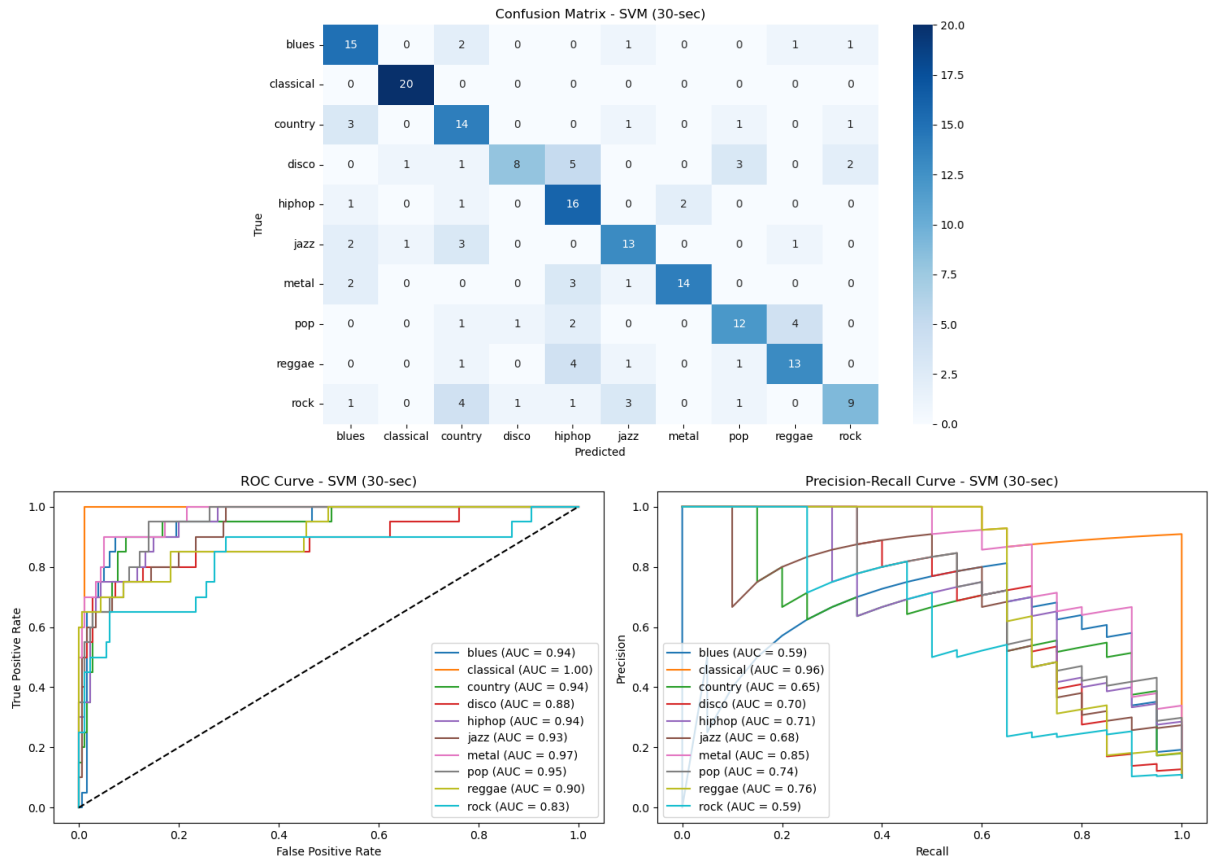


Figure 5: SVM results on 30-second features. Strong genre distinction with high F1 and AUC scores across all classes.

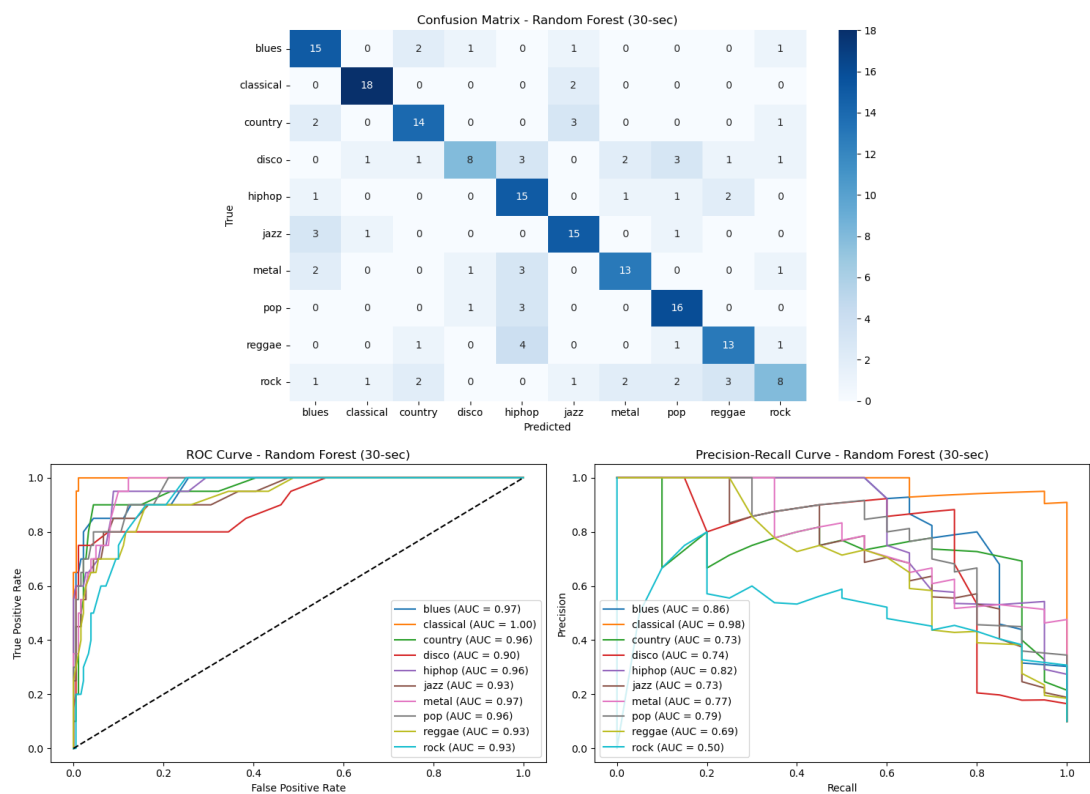


Figure 6: Random Forest on 30-second features. Stable performance and clear separation in evaluation curves.

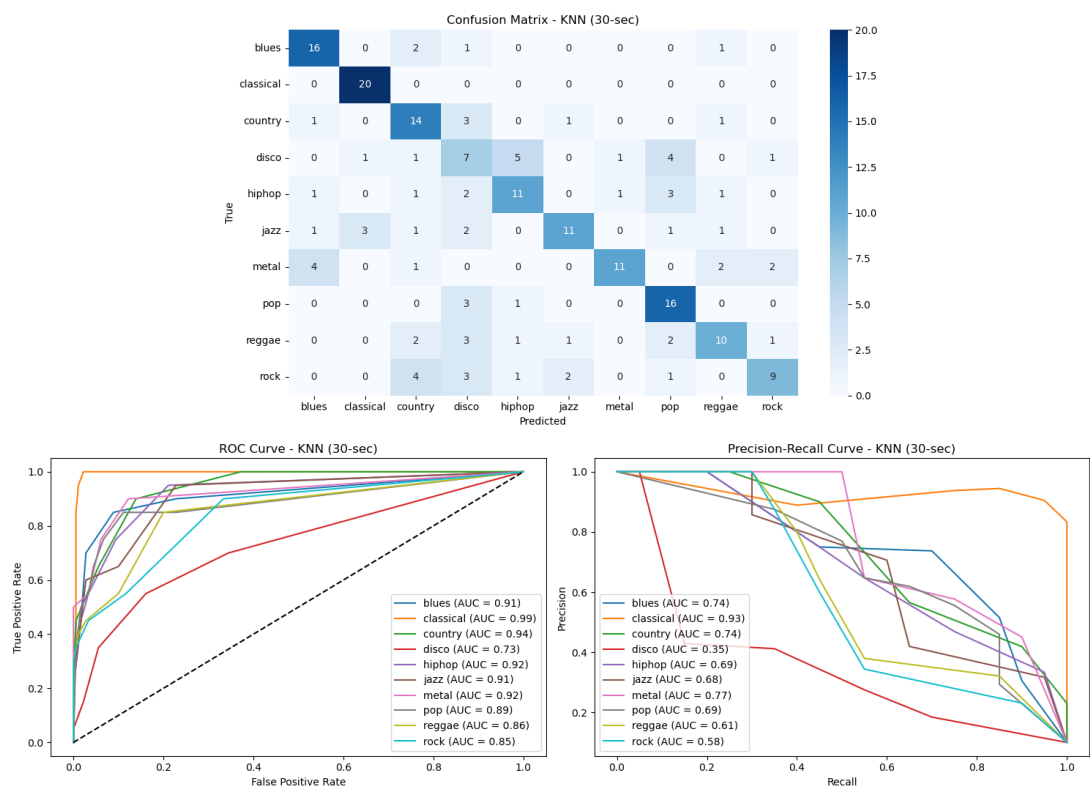


Figure 7: KNN classification on 30-second features. Improved over 3-second input, but some classes remain difficult.

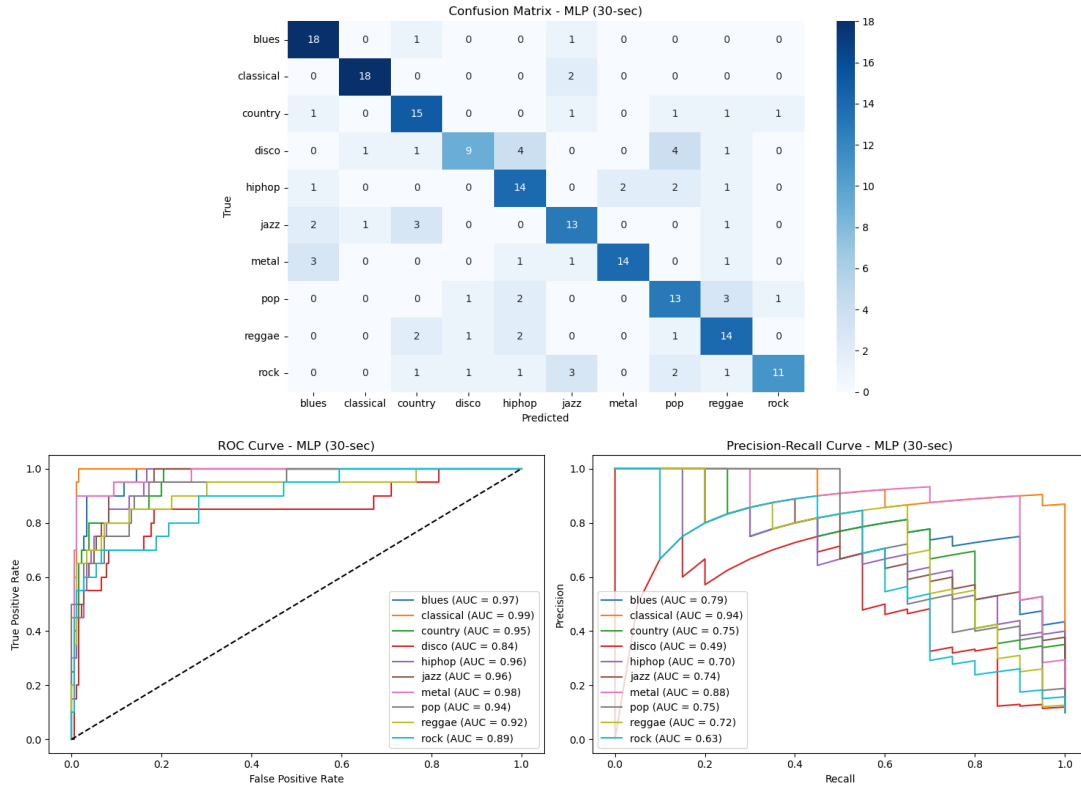


Figure 8: MLPClassifier on 30-second dataset. High accuracy without tuning; curves confirm consistent predictions.

For the spectrogram image dataset, performance was predictably lower relative to feature-based datasets. Prior to classification, spectrograms were converted into grayscale vectors—a preprocessing step that discarded spatial locality and limited the potential for capturing feature dependencies often exploited by deep learning approaches such as Convolutional Neural Networks (CNNs). Despite these limitations, both SVM and MLP achieved reasonable performance, with MLP slightly outperforming SVM (F1-scores of 0.74 and 0.72, respectively). Although these scores were lower than those obtained on engineered feature datasets, they highlight the ability of classical machine learning approaches to extract genre-relational information from spectral patterns despite their suboptimal representation. Confusion matrix analysis revealed stronger misclassification tendencies for styles with overlapping frequency profiles, emphasizing the inherent challenges in this modality.

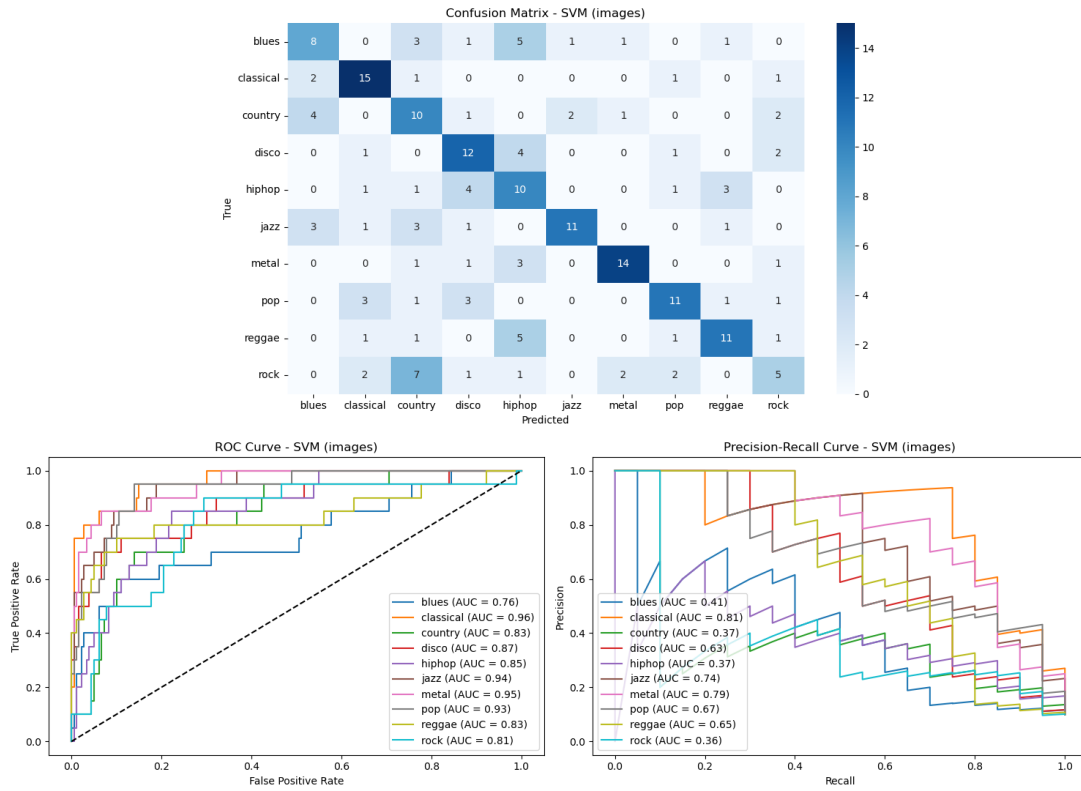


Figure 9: SVM on spectrogram images. Decent performance despite flattened input; curves show mixed confidence.

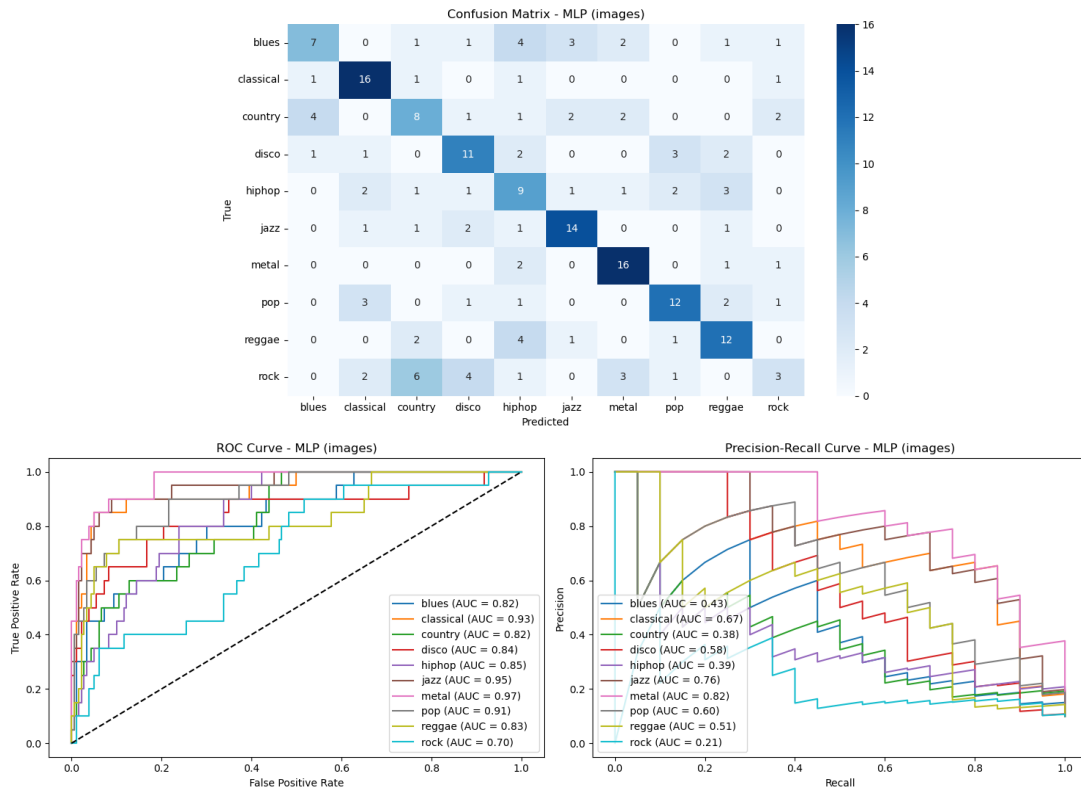


Figure 10: MLP on spectrograms. Slightly better than SVM; good recall in several genres despite spatial loss.

The comparative analysis of model performance across datasets underscored the superiority of engineered audio features over spectrogram images as input representations for classification tasks. Encoded features such as Mel-Frequency Cepstral Coefficients (MFCCs) and chroma profiles offered a streamlined and informative abstraction of relevant acoustic characteristics. Furthermore, the use of 30-second feature representations significantly boosted classification accuracy compared to their 3-second counterparts, reinforcing the importance of temporal context in representing musical structure.

Across all datasets and tasks, the SVM emerged as the most effective classifier. Its ability to generalize well in high-dimensional spaces, coupled with its robustness against overfitting, rendered it particularly suited for tasks involving both short and long audio segment features. Notably, MLPClassifier performed competitively even without hyperparameter tuning, confirming its viability as a robust baseline classifier when data preprocessing is handled effectively.

By contrast, KNN consistently underperformed across all datasets. This aligns with its inherent limitations in high-dimensional spaces, specifically its susceptibility to noise and correlated features despite prior standardization efforts. The model’s reliance on distance-based metrics further exacerbated these issues.

Image classification, although demonstrating relatively lower accuracy, yielded results of significant relevance. Both Support Vector Machines (SVM) and Multi-Layer Perceptrons (MLP) exhibited the capability to extract genre-specific information from the spectral patterns effectively. However, the absence of spatial modeling posed significant limitations to the models’ ability to distinguish between closely related genres. By flattening the image input, the temporal and frequency locality—essential for identifying harmonic patterns and rhythmic textures—was consequently lost.

The analysis of confusion matrices provided further insights into genre classification performance. Genres such as classical, metal, and pop were consistently well-recognized across all datasets, likely due to their distinct spectral and rhythmic attributes. Conversely, genres like jazz, blues, and disco were frequently misclassified—an outcome that can be attributed to their overlapping instrumentation and structural features.

Overall, the findings underscore several key conclusions. First, the representation of data is critically influential in determining model performance. When utilizing classical machine learning techniques, structured audio features emerge as more effective for genre classification. Second, increasing the temporal length of audio input—from 3 seconds to 30 seconds—significantly enhances classification accuracy. Third, among the algorithms tested, the support vector machine (SVM) proved to be the most reliable and consistently accurate classifier. Meanwhile, the multilayer perceptron classifier (MLPClassifier) demonstrated considerable promise as a versatile and robust model, even without extensive parameter tuning. Finally, while spectrogram-based approaches to classification are viable, their efficacy could be greatly improved through the incorporation of convolutional neural network architectures, an exploration beyond the scope of this study.

Dataset	Classifier	F1 Score	Avg. ROC AUC	Notes
3-sec features	SVM	0.80	~0.91	Best performer for short segments
	Random Forest	0.78	~0.88	Slightly less stable across classes
	KNN	0.75	~0.85	Sensitive to noise and class overlap
	MLPClassifier	0.76	~0.89	Competitive despite no tuning
30-sec features	SVM	0.84	~0.93	Best overall model across all datasets
	Random Forest	0.81	~0.90	Strong generalization
	KNN	0.78	~0.86	Improved over 3-sec but still limited
	MLPClassifier	0.82	~0.91	Consistent and reliable
Spectrograms	SVM	0.72	~0.83	Limited by lack of spatial representation
	MLPClassifier	0.74	~0.85	Slightly better than SVM for images

Table 1: summary of model performance across datasets. F1-scores and AUCs are approximate, based on average values over all classes. MLPClassifier was used with default parameters, while the other models were tuned via grid search.

5 Conclusions

This research project examined the application of classical machine learning models to the task of music genre classification, employing two distinct data representations: structured audio features and spectrogram images. A rigorous methodological framework was adopted, encompassing model training, hyperparameter optimization, and a comprehensive evaluation conducted across multiple datasets. The investigation placed particular emphasis on understanding the influence of input data representation, model architecture, and segment duration on classification performance.

The findings affirm that the quality and structure of input data significantly influence the performance of classification models. Specifically, the dataset using 30-second audio segments consistently surpassed the performance of the 3-second dataset, underscoring the importance of longer segments in capturing richer temporal and harmonic contexts necessary for effective genre differentiation. This observation underscores the criticality of adequate temporal resolution in music classification tasks, where the stylistic and structural attributes often unfold over extended periods.

Among the algorithms evaluated, Support Vector Machines (SVM) demonstrated superior consistency and accuracy across both feature datasets. The ability of SVM to manage high-dimensional input spaces, coupled with its resistance to overfitting, rendered it particularly effective for the structured audio features analyzed in this study. Similarly, Random Forest achieved commendable outcomes but fell slightly short of SVM's performance. By contrast, the K-Nearest Neighbors model encountered notable challenges, particularly when processing the more noisy 3-second feature dataset.

The Multi-layer Perceptron (MLPClassifier), employed in this study without extensive hyperparameter tuning, exhibited robust performance across all tested datasets. In numerous instances, its accuracy was comparable to that of meticulously tuned SVM and Random Forest models. These results indicate that neural networks can serve as effective baseline classifiers, provided input data undergoes proper preprocessing and scaling, even when models are utilized as black boxes.

For the spectrogram image datasets, experiments illuminated both opportunities and challenges associated with utilizing classical machine learning techniques for image-based classification. Despite inherent limitations such as spatial relationships being disregarded after flattening image data, both SVM and MLP models effectively extracted genre-related features from the spectrograms. However, the lower F1-scores and AUC metrics achieved with image datasets, compared to structured audio features, indicate that spectrogram classification would be substantially enhanced by convolutional architectures better suited to discerning local time-frequency patterns.

An important insight derived from this study is that genre classification difficulty varies significantly across genres. Classification models consistently performed well in identifying distinct genres such as classical, metal, and pop due to their pronounced spectral, rhythmic, or dynamic characteristics. Conversely, genres like jazz, blues, and disco proved more challenging to classify accurately owing to their shared instrumentation and subtler stylistic elements.

The overall findings underscore the utility of well-engineered audio features for music genre classification, particularly when combined with high-performing algorithms such as SVM capable of leveraging structured high-dimensional data effectively. Moreover, the versatility displayed by neural networks like MLPClassifier highlights their potential for achieving strong baseline results without extensive hyperparameter optimization. While spectrogram-based techniques showed promise, future advancements are likely to benefit from convolutional neural networks (CNNs) capable of fully exploiting spatial dependencies inherent in spectrogram representations.

Future extensions of this research could leverage sophisticated neural architectures—such as CNNs or hybrid models combining audio features with image data—to further enhance performance. The integration of multi-modal approaches may also be explored to capitalize on complementary information derived from both structured audio features and spectrogram images. Additionally, investigations into real-time applications—such as music recommendation systems or live audio tagging—could bridge the gap between research insights and their practical deployment in real-world systems.